Equivalent Circuit Models of Near-Field Coupling between Electrically Small Antennas and TEM Cells

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Abstract. *Bond wires are electrically small structures that provide operating current to light sources in optical sensor packages of ICs. They inevitably form loops, emitting unwanted radiation. A fully enclosing shield from electromagnetic waves is not possible due to the need of apertures for light transmission. Transverse-electromagnetic (TEM) cells are standardized environments for characterizing radiated emissions. This paper develops equivalent circuit models (EQCs) for the near-field coupling between electrically small antennas and TEM cells. The EQC component values are extracted from finite element method (FEM) simulations. The coupling behavior of the EQC is compared with FEM simulations across a frequency range of 1 MHz to 3 GHz, demonstrating agreement within ±2 % and confirming that the proposed models correctly represent the coupling mechanisms.*

Keywords

Electromagnetic Compatibility, TEM Cell, Equivalent Circuit Model, Dipole Moments, Near-Field Coupling.

1. Introduction

Vertical cavity surface emitting lasers (VCSELs) and light emitting diodes (LEDs) are widely used light sources in optical sensor ICs, where bond wires conduct the operating current and form loops that increase electromagnetic interference (EMI). Since the apertures required for light transmission prevent a complete shield enclosure, an alternative strategy is to exploit geometric cancellation effects in the bond wire geometry. This requires a fundamental understanding of how electrically short structures couple to the standardized IC stripline test environment [1]. Because the IC stripline operates on the same TEM wave-propagation principle as a generic TEM cell [2], a rectangular TEM cell is used as a simplified environment for FEM investigations. Since the bond wires are electrically small, their radiation is fully characterized by equivalent electric and magnetic dipole moments. The coupling paths are predominantly quasi-static, thereby allowing them to be modelled by lumped circuit elements. The resulting EQC provides physical insight into the coupling mechanisms and is simulated orders of magnitude faster than the respective FEM model.

2. Dipole Moments in TEM Cells

The TEM cell provides a controlled environment for characterizing EMI of a device-under-test (DUT) [2]. The septum acts as a transmission line with wave impedance $Z_w = 50\,\Omega$. The DUT is placed between the septum and the cell enclosure as shown in Fig. 1. From the terminal voltages U_1 and U_2, the magnitudes of the electric and magnetic dipole moments, $|m_e|$ and $|m_m|$, are derived as [3]

$$|m_e| = \frac{|U_1 + U_2|h}{Z_w}, \quad |m_m| = \frac{|U_1 - U_2|hZ_0}{Z_w} \tag{1}$$

where Z_0 denotes the impedance of free space and h is the distance from the septum to the DUT. The DUT is the IC package including the bond wires, for which these dipole moments are derived.

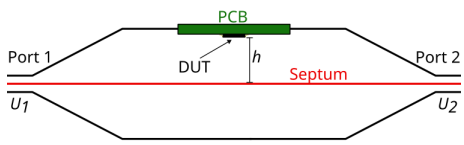

Fig. 1. TEM cell with DUT and PCB.

3. Equivalent Circuit Models

Electrically small antennas are modelled by an RLC circuit: C_A represents the stored electric energy, L_A stored magnetic energy, and R_A radiation losses. Ohmic losses are neglected by using perfect electric conductors, focusing the analysis purely on the coupling behaviour. The circuit parameters are extracted by placing the antenna on a sufficently large PEC ground plane in free space, using stored energy integrals from FEM simulations. The TEM cell is modelled as a symmetric transmission line ($L_{T,1}$, $C_{T,1}$, $L_{T,2}$, $C_{T,2}$), terminated with $R_1 = R_2 = 50\,\Omega$. It is connected to the

antenna model with a coupling capacitor C_k and mutually coupled inductances ($L_A \leftrightarrow L_{T,1}, L_{T,2}$).

3.1. Loop Antenna

The loop antenna is modelled by a parallel L_A and C_A connection, accurately representing inductive behaviour at low frequencies and increasingly capacitive behaviour with rising frequency. The radiation resistance R_A is neglected, as radiation is fully accounted for by the coupling components introduced. The full circuit of this configuration is shown in Fig. 2. The dipole moments are derived from

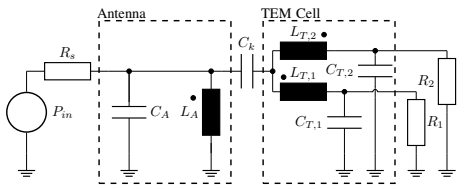

Fig. 2. EQC of the loop antenna and TEM cell. Polarity of mutual inductances between L_A and $L_{T,1}$, $L_{T,2}$ indicated by dots next to inductor symbols.

the terminal voltages U_1, U_2, decomposed into contributions from capacitive coupling C_k and mutual inductance coupling, as shown in Fig. 3. It is found that the electric dipole moment arises solely from the displacement current through C_k, and the magnetic dipole moment solely from the voltages induced across $L_{T,1}$ and $L_{T,2}$.

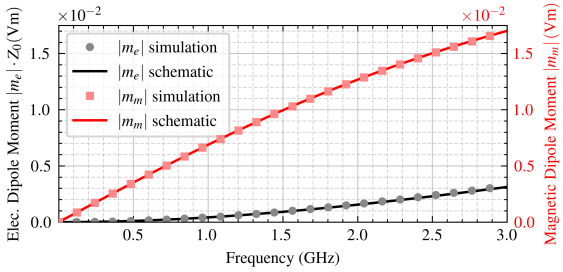

Fig. 3. Equivalent electric and magnetic dipole moments of the loop antenna derived from FEM simulations of the TEM cell. The electric dipole moment m_e is weighted with Z_0 for comparison.

3.2. Monopole Antenna

The monopole antenna in a TEM cell is investigated next, to verify the EQC concept for a predominantly capacitive antenna. The monopole EQC follows the same topology as Fig. 2, except that C_A is connected in series with L_A, accurately modelling the capacitive behaviour in low frequencies and increasingly inductive behaviour with rising frequency. The coupling capacitance C_k is connected at the node between C_A and L_A. The derived dipole moments are shown in Fig. 4.

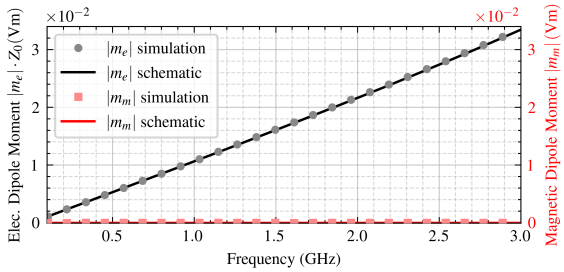

Fig. 4. Equivalent electric and magnetic dipole moments of the monopole antenna derived from FEM simulations of the TEM cell. The electric dipole moment m_e is weighted with Z_0 for comparison.

4. Conclusion

The EQCs presented accurately model the coupling behaviour of the loop and the monopole antennas to the TEM cell. The simulation is orders of magnitude faster than FEM. The EQC reveals that the equivalent electric dipole moment is directly associated with capacitive coupling to the TEM cell septum via C_k, while the magnetic dipole moment arises from inductive coupling between L_A and the septum inductances. Further research could extend the models to higher-order modes, shielding materials, or other electrically small antenna geometries.

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About Authors…

Simon PRATO was born in Vöcklabruck, Austria, in 2002. He received an M.Sc. degree from the Graz University of Technology in 2026. He was employed as a student worker at Infineon Technologies from 2022 to 2025 and was also involved in teaching undergraduate mathematics courses.